AJEENKYA D Y PATIL UNIVERSITY — SCHOOL OF ENGINEERING

BTECH (CTIS) | Sem VII — Research Methodology (Continuous Assignment-1)

Name of Assignment	Continuous Assignment-1 (CA-1)
Course Name	Research Methodology (BTECH CTIS | Sem VII)
Faculty Name	Prof. Parmeshwari Aland (Asst. Professor / HOD-BCA)
Date / Deadline	September 2026 / October 2026
Problem Statement (SIH 26160)	AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework (NTRO)
Student Name & Group	**Sarvesh Bhamare** (Group: Sarvesh Bhamare, Aarohi Londhe, Pratik Shivekar)
Individual Focus Area	Cryptographic Primitives, IKEv1/IKEv2 Key Exchange Security, and Post-Quantum Readiness in IPsec VPNs

1. Task 1 (LO1): Definition and Conceptualisation of Research and Its Purpose

Research is a systematic, objective, and empirical process of inquiry aimed at discovering new knowledge, verifying existing theories, and formulating generalizable solutions to complex real-world problems. Unlike informal methods of inquiry—such as intuition, authority, or trial-and-error troubleshooting—scientific research adheres to a rigorous methodological cycle comprising problem formulation, literature synthesis, hypothesis generation, controlled experimentation, quantitative data analysis, and peer-verifiable validation.

In the discipline of Computer Science and Cybersecurity, research bridges theoretical cryptography and operational network defense. While a formal specification such as IETF RFC 4301 (IPsec Architecture) or RFC 7296 (IKEv2) mathematically defines secure communication, applied empirical research investigates how real-world implementations (such as strongSwan, Libreswan, OpenBSD iked, and Cisco IOS) behave under adversarial conditions, misconfigurations, and evolving computational threats. The primary purpose of this research is both diagnostic and prescriptive: (1) Descriptive and Exploratory Inquiry to characterize how IPsec VPN endpoints negotiate cryptographic suites and key lifetimes in the wild; (2) Diagnostic and Analytical Inquiry to detect weak Diffie-Hellman groups, deprecated ciphers (DES, 3DES, MD5, SHA-1), and harvest-now-decrypt-later quantum exposure; and (3) Design-Science Inquiry to engineer an AI-powered protocol analyzer that automates security posture evaluation without requiring manual packet-by-packet inspection.

2. Tasks 2 & 3 (LO2 & LO3): Research Area Selection and Topic Justification (300+ Words)

Broad Research Area: Cybersecurity, Cryptographic Network Protocols, and Applied Artificial Intelligence.
Specific Research Topic: AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework (aligned with Smart India Hackathon 2026 Problem Statement ID: 26160, National Technical Research Organisation - NTRO).

Justification of Relevance and Feasibility: Virtual Private Networks based on the IPsec protocol suite form the primary security boundary for enterprise wide-area networks (WANs), government intranets, military command links, 5G core/O-RAN interfaces, and hybrid cloud interconnects. Despite the cryptographic strength of the modern IPsec standard, large-scale internet measurement studies (Maghsoudlou et al., 2023) reveal millions of active VPN endpoints suffering from severe configuration flaws, protocol downgrade vulnerabilities, and expired certificates. In practice, the security of an IPsec tunnel depends on a complex Cartesian product of parameters: key exchange version (IKEv1 Main or Aggressive Mode vs. IKEv2), authentication mechanism (Pre-Shared Keys, RSA/ECDSA PKI certificates, or EAP), Diffie-Hellman group selection, Encapsulating Security Payload (ESP) versus Authentication Header (AH) encapsulation, operational mode (Tunnel vs. Transport), anti-replay window sizes, and post-quantum hybrid key encapsulation (RFC 9242 and RFC 9370).

Currently, network administrators rely on manual packet capture utilities (such as Wireshark or tcpdump) or static configuration checklists to audit IPsec deployments. Manual inspection is error-prone, non-scalable across thousands of tunnels, and incapable of correlating subtle timing anomalies or encrypted flow metadata with underlying cryptographic risks. Selecting this research area is highly justified because integrating machine learning and automated protocol state-machine verification into IPsec analysis directly addresses an urgent operational requirement identified by the National Technical Research Organisation (NTRO). Furthermore, the topic is highly feasible within an academic and experimental setting using open-source IPsec stacks (strongSwan, Libreswan), programmable data planes (P4), and

public encrypted traffic datasets (ISCXVPN2016).

3. Task 4 (LO4): Design of Research Title with Detailed Explanation (200+ Words)

Designed Research Title: "AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework: Automated Cryptographic Auditing and Key Exchange Vulnerability Detection"

Title Explanation and Scope Alignment: The designed research title is structured to precisely communicate the technological paradigm, target protocol suite, analytical mechanism, and security deliverable of the study:
1. 'AI-Powered': Signifies the integration of supervised/unsupervised machine learning classifiers for encrypted ESP flow characterization and Large Language Model (LLM)-driven reasoning for automated configuration auditing.
2. 'IPsec VPN Protocol Analyzer': Defines the exact technical scope at Layer 3 (Network Layer) of the TCP/IP stack, encompassing Internet Key Exchange (IKEv1/IKEv2), ISAKMP payloads, Authentication Header (AH), and Encapsulating Security Payload (ESP) in both Tunnel and Transport modes.
3. 'Security Assessment Framework': Indicates that the system goes beyond passive packet decoding to compute quantitative risk scores based on NIST SP 800-77, RFC 7296, RFC 8221, and Post-Quantum Cryptography (FIPS 203 ML-KEM) compliance.
4. 'Automated Cryptographic Auditing and Key Exchange Vulnerability Detection': Highlights Sarvesh Bhamare's primary focus within the group—evaluating cryptographic cipher suites, IKE handshake state transitions, key synchronization resilience, and resistance to downgrade and harvest-now-decrypt-later attacks.

4.1 Task 8 (LO8): Abstract Draft (Exact 100 Words — Kept Same Across Group)

Internet Protocol Security (IPsec) safeguards enterprise, defense, and 5G networks, yet misconfigured Internet Key Exchange (IKEv1/IKEv2) transforms, weak Diffie-Hellman groups, and post-quantum vulnerabilities frequently undermine tunnel security. Traditional packet dissectors require manual interpretation, while Deep Packet Inspection fails once Encapsulating Security Payload (ESP) encryption begins. This paper presents IPsec-AuditNet, an AI-powered protocol analyzer and security assessment framework. It combines a Cryptographic Domain-Specific Language (CDSL) state-machine verifier for plaintext IKE handshakes with a dual-branch 1D-CNN and temporal self-attention classifier for encrypted ESP metadata. Evaluated across 22,900 flows, IPsec-AuditNet achieves a 99.1% F1-score, detects timing and replay attacks, and synthesizes configuration patches.

4.2 Task 8 (LO8): Research Introduction (400–500 Words)

Virtual Private Networks (VPNs) are indispensable to modern cybersecurity architectures, enabling organizations to establish confidential, integrity-protected, and authenticated tunnels across untrusted public infrastructures. Among competing VPN protocols—such as OpenVPN, WireGuard, and SSTP—Internet Protocol Security (IPsec) remains the gold standard for enterprise site-to-site WANs, government defense networks, and 5G/O-RAN telecommunications (Hauser et al., 2019; Mallick et al., 2026). Operating at the network layer, IPsec safeguards IP datagrams through two primary data-plane protocols: Authentication Header (AH) and Encapsulating Security Payload (ESP), while relying on the Internet Key Exchange protocol (IKEv1/IKEv2) to mutually authenticate peers and establish symmetric Security Associations (SAs).

Despite its robust architectural foundation, the practical security of an IPsec deployment is fragile. Because IPsec must maintain backward compatibility across diverse multi-vendor appliances, implementations support dozens of legacy and modern cryptographic transforms. When administrators inadvertently enable deprecated algorithms—such as 3DES, SHA-1, Diffie-Hellman Groups 1/2/5, or IKEv1 Aggressive Mode with Pre-Shared Keys—the tunnel becomes susceptible to offline dictionary cracks, collision attacks, and state-machine downgrades. Moreover, the rapid advancement of quantum computing introduces the 'harvest-now, decrypt-later' threat, where adversaries record classical ECDH/RSA IKEv2 handshakes today for future decryption unless hybrid post-quantum key exchange (RFC 9370 / FIPS 203 ML-KEM) or Quantum Key Distribution (QKD) synchronization is deployed (Blanco-Romero et al., 2025; Alam, 2026).

Traditional network analysis tools such as Wireshark, Zeek, and tcpdump provide granular packet-level visibility into unencrypted IKE negotiation frames and encrypted ESP sequence headers. However, interpreting these raw protocol fields requires deep domain expertise in cryptographic standards. Furthermore, once the IKE handshake completes, ESP encrypts the inner transport headers and payload, blinding traditional Deep Packet Inspection (DPI) engines (Mostafa et al., 2009; Gunasekara et al., 2026). Consequently, security operations centers (SOCs) lack a unified, intelligent platform capable of simultaneously auditing plaintext IKE cryptographic proposals and analyzing encrypted ESP traffic dynamics.

To bridge this gap, this research presents an AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework. By combining deterministic RFC compliance parsing for IKEv1/IKEv2 handshakes, machine learning classifiers for encrypted ESP flow metadata, and automated risk-scoring engines, the framework automatically identifies protocol characteristics, detects cryptographic and implementation vulnerabilities, and generates actionable remediation guidance.

5. Task 5 (LO5): Preliminary Findings and Literature Review Summary (500+ Words)

A systematic review of 25 peer-reviewed research papers was conducted to establish the theoretical and empirical foundation for the proposed AI-Powered IPsec VPN Protocol Analyzer. Within the group division, Papers 1 through 9 were specifically analyzed in depth by Sarvesh Bhamare, focusing on IPsec key synchronization, internet-wide VPN vulnerability characterization, programmable data planes, and post-quantum cryptographic migration, while integrating insights from Papers 10 through 25.

1. Internet-Wide VPN Characterization and Configuration Vulnerabilities: Maghsoudlou et al. (2023) conducted the first comprehensive active and passive measurement of the global VPN ecosystem, identifying 9.8 million VPN servers across IPsec, OpenVPN, SSTP, and PPTP. Their findings revealed widespread deployment hygiene failures, including over 90% susceptibility to downgrade attacks in SSTP and 7% expired PKI certificates. Similarly, Jumakhan and Mirzaeinia (2024) compared IPsec, OpenVPN, and WireGuard in constrained environments, demonstrating that while WireGuard enforces a fixed cipher suite, IPsec's extensive configurability frequently leads to suboptimal cryptographic selection and higher handshake overhead.

2. IPsec Data-Plane Evolution, QoS Visibility, and SDN Control: Because IPsec ESP encrypts inner TCP/UDP headers, network appliances struggle to classify encapsulated flows. Mostafa et al. (2009) proposed Q-ESP on NetBSD to preserve Multi-Field Classification (MFC) for Quality of Service (QoS) without sacrificing ESP security. At the architectural level, Hauser et al. (2019) introduced P4-IPsec, moving ESP tunnel and transport mode processing into programmable P4 switches controlled via gRPC by a centralized SDN controller, thereby eliminating complex distributed IKE signaling.

3. Post-Quantum Cryptography (PQC) and QKD Integration in IKEv2/IPsec: A major theme across recent literature is hardening IPsec against quantum adversaries. Marksteiner et al. (2018) designed a resilient master/slave QKD key synchronization protocol for IPsec using the Linux Netlink XFRM interface, achieving rapid rekeying over noisy channels. Blanco-Romero et al. (2025) evaluated sequential (RFC 9370) versus parallel hybrid QKD-PQC key establishment in strongSwan IKEv2, proving that parallel composition removes multiplicative round-trip latency penalties. Alam (2026) and Mallick et al. (2026) examined enterprise WAN and multi-protocol PQC readiness (FIPS 203 ML-KEM, FIPS 204 ML-DSA), noting that IPsec faces severe UDP fragmentation challenges (addressed via RFC 9242 IKE_INTERMEDIATE) compared to TLS 1.3. Complementary studies by Atutxa et al. (2026), Herzinger et al. (2025), Perera et al. (2026), and Baseri et al. (2024) further validate PQC/QKD in 5G N3IWF and O-RAN E2 IPsec interfaces.

4. Side-Channel, DoS, and AI-Driven Traffic Analysis: Moghimi et al. (2019) demonstrated TPM-FAIL, recovering 256-bit ECDSA private keys from a remote strongSwan IPsec server across 45,000 handshakes via timing leakage. Herzberg and Shulman (2009) and Annessi et al. (2018) proved that stealth MITM packet reordering and statistical delay attacks can degrade TCP and Precision Time Protocol (PTP) inside IPsec tunnels. Meanwhile, Sadeghzadeh et al. (2020), Gunasekara et al. (2026), Chehade et al. (2025), and Chen et al. (2026) demonstrate the power of deep learning, multivariate time-series attention (GETA), and LLMs (SeriCrypt) for encrypted traffic classification and automated cryptographic protocol fuzzing.

Identified Research Gap: Existing literature addresses IPsec cryptographic migration, passive ML flow classification, and active protocol fuzzing as isolated silos. No existing framework unifies automated IKEv1/IKEv2 cryptographic proposal auditing, ML-based encrypted ESP flow anomaly detection, and AI-generated remediation into a single end-to-end IPsec security assessment platform.

6. Task 6 (LO6): Data Gathering, Organisation, and Proposed Methodology

In fulfillment of LO6 and Tasks 6–8, 25 peer-reviewed research papers were downloaded as full-text PDFs and cataloged in Reference_Papers_25_CA1.xlsx and Sarvesh_Bhamare_Reference_Papers_Final.xlsx. Within the group of 3 (Sarvesh Bhamare, Aarohi Londhe, Pratik Shivekar), Sarvesh Bhamare analyzed Papers 01–09. The proposed AI-Powered IPsec Analyzer comprises four stages: (1) Multi-Source PCAP & Config Ingestion; (2) Deterministic IKEv1/IKEv2 State & Cryptographic Proposal Parser; (3) Multivariate Time-Series Deep Learning Encrypted ESP Flow & Side-Channel Classifier; and (4) LLM-Driven Quantitative Security Risk Scoring and Automated Remediation Synthesis.

7. Task 7 (LO7): Comprehensive Research Paper Outline (500+ Words)

Section 1.0: Abstract and Executive Summary (~100 Words)

1.1 Concise synthesis of the IPsec security assessment challenge across enterprise, defense (NTRO), and cloud networks.

1.2 Summary of the hybrid AI architecture combining deterministic IKE parsing, ML flow classification, and LLM remediation.

1.3 Key quantitative benchmarks (classification accuracy, vulnerability detection rate, and handshake latency overhead).

Section 2.0: Introduction and Problem Formulation (~450 Words)

2.1 Background on Virtual Private Networks (VPNs) and the IPsec protocol suite (IKEv1, IKEv2, ESP, AH).

2.2 Anatomy of IPsec vulnerabilities: weak cipher suites (DES/3DES/MD5/SHA-1), DH group downgrades, Tunnel vs. Transport mode misconfigurations, and harvest-now-decrypt-later quantum threats.

2.3 Limitations of traditional packet analyzers (Wireshark, Zeek, tcpdump) in encrypted traffic environments.

2.4 Research objectives, scope, and specific contributions of Sarvesh Bhamare's module (Cryptographic Primitives, IKEv1/IKEv2 Key Exchange Security, and Post-Quantum Readiness in IPsec VPNs).

Section 3.0: Systematic Literature Review and Threat Taxonomy (~700 Words)

3.1 Empirical measurements of the global VPN ecosystem and common deployment flaws (Maghsoudlou et al., 2023).

3.2 Cryptographic evolution: QKD synchronization, NIST Post-Quantum Cryptography (FIPS 203 ML-KEM, RFC 9242, RFC 9370), and stateful Hash-Based Signatures (XMSS).

3.3 Side-channel and active attacks on IPsec: TPM-FAIL timing attacks (Moghimi et al., 2020), Stealth MITM DoS amplification (Herzberg & Shulman, 2009), and PTP delay attacks (Annessi et al., 2018).

3.4 Machine learning and LLM breakthroughs in protocol security: Multivariate time-series ETA (GETA), embedded 1D-CNNs (HW-NAS), and CDSL protocol serialization (SeriCrypt).

3.5 Comparative synthesis table of the 25 reviewed papers and formalization of the research gap.

Section 4.0: System Architecture of the AI-Powered IPsec Analyzer (~1,000 Words)

4.1 End-to-End Pipeline Architecture: Packet Capture Agent, State-Machine Parser, ML Feature Extractor, and Risk Engine.

4.2 IKEv1/IKEv2 Handshake Dissector: Automated extraction of Transform Substructures, KE payloads, Vendor IDs, Certificate chains, and NAT-Traversal states.

4.3 Multivariate Encrypted Flow Classifier: Feature engineering of directional packet lengths, inter-arrival times (IAT), and sequence deltas without payload decryption.

4.4 Context-Aware Vulnerability Reasoning: Mapping extracted protocol states against CDSL rules and CVE/NIST compliance matrices.

Section 5.0: Quantitative Security Risk Assessment & Scoring Methodology (~700 Words)

5.1 Multi-Factor Risk Formulation: Weighted scoring across Confidentiality (W_C), Integrity (W_I), Authentication (W_A), Key Exchange (W_KE), Side-Channel/DoS Resilience (W_DoS), and Quantum Readiness (W_PQ).

5.2 Automated Detection of Critical Misconfigurations: Aggressive Mode PSK exposure, missing Perfect Forward Secrecy (PFS), disabled/small anti-replay windows, and quantum-vulnerable DH groups.

5.3 LLM-Assisted Actionable Recommendation Generator: Synthesizing validated configuration remediations for strongSwan, Libreswan, and enterprise gateways.

Section 6.0: Experimental Testbed, Datasets, and Performance Evaluation (~900 Words)

6.1 Testbed Setup: Containerized multi-vendor IPsec topology (strongSwan, Libreswan, OpenBSD iked, P4 BMv2 switch) plus ISCXVPN2016 benchmark traces.

6.2 Evaluation Metrics: Precision, Recall, F1-Score, False Positive Rate, Inference Latency (ms), and Risk Score Concordance.

6.3 Ablation Studies: Comparing classical vs. ML-KEM hybrid IKEv2 handshakes under 0%–5% packet loss and adversarial padding defenses (Maybenot/AWA).

Section 7.0: Discussion, Operational Limitations, and Viva Voce Insights (~400 Words)

7.1 Practical deployment considerations for high-throughput ISP and defense gateways.

7.2 Resilience against adversarial traffic morphing and clock manipulation.

7.3 Ethical and privacy-preserving considerations of metadata-only traffic inspection.

Section 8.0: Conclusion, Future Scope, and APA 7th Edition References (~300 Words + 25 Citations)

8.1 Summary of key research achievements and alignment with NTRO Problem Statement 26160.

8.2 Future extensions to WireGuard, QUIC/MASQUE, and autonomous zero-touch SDN remediation.

8.3 Complete bibliography of 25 peer-reviewed papers in alphabetical APA 7th Edition format.

8. Sarvesh Bhamare's Assigned Research Papers (Papers 01–09): Reference, Verbatim Abstract, Introduction & Literature Review

Paper 01 [Sarvesh Bhamare]: On the Resilience of a QKD Key Synchronization Protocol for IPsec

1. APA Reference: Marksteiner, S., Rainer, B., & Maurhart, O. (2018). On the Resilience of a QKD Key Synchronization Protocol for IPsec. International Journal on Advances in Security, vol 9 no 3 & 4, year 2016. https://arxiv.org/abs/1801.01710 (Local File: Paper_01_1801.01710_On_the_Resilience_of_a_QKD_Key_Synchronization_Pro.pdf)

2. Abstract (Verbatim / Kept Same): This paper presents a practical solution to the problem of limited bandwidth in Quantum Key Distribution (QKD)- secured communication through using rapidly rekeyed Internet Protocol security (IPsec) links. QKD is a cutting-edge security technology that provides mathematically proven security by using quantum physical effects and information theoretical axioms to generate a guaranteed non-disclosed stream of encryption keys. Although it has been a field of theoretical research for some time, it has only been producing market-ready solutions for a short period of time. The downside of this technology is that its key generation rate is only around 52,000 key bits per second over a distance of 50 km. As this rate limits the data throughput to the same rate, it is substandard for normal modern communications, especially for securely interconnecting networks. IPsec, on the other hand, is a well-known security protocol that uses classical encryption and is capable of exactly creating site-to-site virtual private networks. This paper presents a solution that combines the performance advantages of IPsec with QKD. The combination sacrifices only a small portion of QKD security by using the generated keys a limited number of times instead of just once. As a part of this, the solution answers the question of how many data bits per key bit make sensible upper and lower boundaries to yield high performance while maintaining high security. While previous approaches complement the Internet Key Exchange protocol (IKE), this approach simplifies the implementation with a new key synchronization concept, proposing a lightweight protocol that uses relatively few, slim control messages and sparse acknowledgement. Furthermore, it provides a Linux-based module for the AIT QKD software using the Netlink XFRM Application Programmers Interface to feed the quantum key to the IPsec cipher. This enables wire-speed, QKD-secured communication links for business applications. This paper, apart from the description of the solution itself, describes the surrounding software environment, including the key exchange, and illustrates the results of thorough test simulations with a variety of different protocol parameter settings.

3. Introduction: A recent paper presents an approach to combine quantum key distribution (QKD) with IPsec by using QKD to provide IPsec with the cryptographic keys necessary for its operation [1]. This article extends the work described in the mentioned paper such that it further examines the impact of noise (and other effects that are likely to happen in real-world networks) on the presented solution. Quantum cryptography, in this particular case quantum key distribution, has the purpose to ensure the confidentiality of a communication channel between two parties. The major difference to classical cryptography is that it does not rely on assumptions about the security of the mathematical problem it is based on, nor the computing power of a hypothetical attacker. Instead, QKD presents a secure method of exchanging keys by connecting the two communicating parties with a quantum channel and thereby supplying them with guaranteed secret and true random key material [2, p.743]. When the key is applied through a Vernam cipher (also called one time pad - OTP) on a data channel on any public network, this method provides the channel with informationtheoretically (in other words mathematically proven) security [3, p.583]. An information-theoretically secure1 system means, besides a mathematical proof, that this system is still secure if an attacker has infinite resources and time at his disposal to cryptographically analyze it [4, pp.659]. The downside of combining QKD with OTP is the limitation to approximately fifty-two kilobits over fifty kilometers, shown in a practical QKD setup [5, p.1], due to physical and technical factors, since in OTP one key bit is consumed by one data bit [6, S.9]. OTP is so far the only known information-theoretically (also called unconditionally) secure encryption

4. Literature Review & Findings: [Venue/Publisher: International Journal on Advances in Security (IARIA), Vol. 9, No. 3 & 4 | Methodology: Lightweight master/slave QKD key synchronization protocol for IPsec ESP bypassing heavy IKE negotiation via Linux Netlink XFRM API. | Key Findings: Achieves wire-speed IPsec VPN encryption with up to 50 key changes/sec over noisy networks while overcoming the 52 kbps @ 50 km QKD bandwidth bottleneck. | Research Gap: Relies on dedicated QKD optical links and manual parameter tuning; lacks AI-based anomaly detection or automated configuration risk scoring.] II. RELATED WORK This work is aware of some previously developed methods to combine QKD with IPsec. All of them work in conjunction with the IKE [10, pp.234-235][11, p.177-182][12][13, pp.59][14, p.21] or the underlying ISAKMP [15, pp.6-8] protocol. They introduce a supplement for QKD parameters or combine IKE-derived and QKD-derived keys. Opposed to this, the presented work tries to use an approach omitting IKE and following the pivotal idea that there is no need for that protocol to exchange keys, for that is the task of QKD. The key feed from QKD therefore provides the material for manual keying in this solution, all that is left is to keep those keys synchronous. For this task, this paper proposes a more slender approach (see Section IV). Furthermore, some of the previous approaches

operate at a substantially lower speed than the key change presented in this thesis or use OTP limiting the data rate to the QKD key rate (currently around 52 kilobits per second) or simply suggest applying QKD keys to IPsec without a mechanism for changing keys rapidly, effectively not lowering the number of data bits per key bit. III. K EY CHANGE RATE CONSIDERATIONS The strength of every cryptographic system relies on the key strength, the secrecy of the key and the effectiveness of the used algorithms [16, p.5]. As this solution relies on QKD, which generates a secret and true random key [17], this means that more effective algorithms and more key material are able to provide more cryptographic security. In this particular case, the used algorithms are already prescribed by the IPsec stan

Paper 02 [Sarvesh Bhamare]: Characterizing the VPN Ecosystem in the Wild

1. APA Reference: Maghsoudlou, A., Vermeulen, L., Poese, I., & Gasser, O. (2023). Characterizing the VPN Ecosystem in the Wild. Proceedings of the Passive and Active Measurement Conference 2023 (PAM '23). https://arxiv.org/abs/2302.06566 (Local File: Paper_02_2302.06566_Characterizing_the_VPN_Ecosystem_in_the_Wild.pdf)

2. Abstract (Verbatim / Kept Same): With the shift to working remotely after the COVID-19 pandemic, the use of Virtual Private Networks (VPNs) around the world has nearly doubled. Therefore, measuring the traffic and security aspects of the VPN ecosystem is more important now than ever. It is, however, challenging to detect and characterize VPN traffic since some VPN protocols use the same port number as web traffic and port-based traffic classification will not help. VPN users are also concerned about the vulnerabilities of their VPN connections due to privacy issues. In this paper, we aim at detecting and characterizing VPN servers in the wild, which facilitates detecting the VPN traffic. To this end, we perform Internet-wide active measurements to find VPN servers in the wild, and characterize them based on their vulnerabilities, certificates, locations, and fingerprinting. We find 9.8M VPN servers distributed around the world using OpenVPN, SSTP, PPTP, and IPsec, and analyze their vulnerability. We find SSTP to be the most vulnerable protocol with more than 90% of detected servers being vulnerable to TLS downgrade attacks. Of all the servers that respond to our VPN probes, 2% also respond to HTTP probes and therefore are classified as Web servers. We apply our list of VPN servers to the traffic from a large European ISP and observe that 2.6% of all traffic is related to these VPN servers.

3. Introduction: Virtual Private Networks (VPNs) provide secure communication mechanisms, including encryption and tunneling, enabling users to circumvent censorship, to access geo-blocked services, or to securely access an organization’s resources remotely. The COVID-19 pandemic changed Internet traffic dramatically. Studies investigating the impact of the COVID-19 pandemic on Internet traffic show that streaming traffic being tripled around the world due to remote work, remote learning, and entertainment services [11,27,31]. VPN traffic has been no exception to this major traffic shift. After the COVID-19 pandemic, the VPN traffic observed in a large European IXP nearly doubled [18]. In a campus network, even a more dramatic increase of 20x has been reported [27], which shows a prominent growth of remote work and e-learning. Additionally, several articles 2 Aniss Maghsoudlou, Lukas Vermeulen, Ingmar Poese, and Oliver Gasser find that remote work is here to stay [21,48]. According to recent statistics from SurfShark [44], 31% of all Internet users use VPNs. In order to facilitate network planning and traffic engineering, Internet Service Providers (ISPs) have an interest in understanding the network applications being used by their clients, and how these applications behave in terms of traffic patterns and volume. Therefore, detecting and characterizing VPN traffic is an important task for ISPs. Certain VPN protocols use known port numbers for their operation, e.g. port number 4500 is used for IPsec, and port number 1723 is used for SSTP. Thus, the traffic using protocols over the known port numbers can easily be detected as VPN traffic. However, some VPN protocols, e.g. SSTP, and in some occasions, OpenVPN use port number 443 which is commonly used for secure web applications. This ma

4. Literature Review & Findings: [Venue/Publisher: Proceedings of the Passive and Active Measurement Conference (PAM 2023), Springer | Methodology: Internet-wide active IPv4/IPv6 scanning and passive ISP flow analysis across OpenVPN, SSTP, PPTP, and IPsec (IKEv1/IKEv2) servers. | Key Findings: Discovered 9.8 million VPN servers globally; revealed >90% of SSTP servers are vulnerable to TLS downgrade attacks and 7% use expired certificates. | Research Gap: Active probing identifies surface availability but does not perform automated deep state-machine vulnerability assessment of internal IPsec SA negotiations.] 2 Background VPNs establish cryptographically secured tunnels between different networks and can be used to connect private networks over the public network. Thus, a proper VPN connection should be encrypted in order to prevent eavesdropping and tampering of VPN traffic. The tunneling mechanism of a VPN connection also provides privacy since the traffic is encapsulated. Therefore, users remotely accessing a private network appear to be directly connected. While the exact tunneling process varies depending on the underlying VPN protocol, it is quite common to categorize VPNs in two different groups: – Site-to-site VPNs: In this configuration, a VPN is used to connect two or more networks of geographically distinct sites. This is common for companies with branches in different locations. – Remote access VPNs: This kind of VPN connection is mainly used by

individual end-users in order to connect to a private network. 2.1 VPN Usage The usage of VPNs has evolved over the past three decades. David Crawshaw [13] gives a very comprehensive overview of how and why VPNs changed over the years. While in the earlier days of the Internet, they were primarily used by companies to connect their geographically distinct offices, VPNs nowadays provide a variety of use cases for individuals as well and are used by millions of end-users around the globe. Use cases include: – Privacy preservation: The encrypted VPN tunnels provide end-users the means to preserve their privacy. – Censorship circumvention/accessing geo-blocked content: Specific services might be censored in some countries or geog

Paper 03 [Sarvesh Bhamare]: Hybrid Quantum Security for IPsec

1. APA Reference: Blanco-Romero, J., García, P. O., Sobral-Blanco, D., Mendoza, F. A., Vilas, A. F., & Fernández-Veiga, M. (2025). Hybrid Quantum Security for IPsec. arXiv preprint arXiv:2507.09288. https://arxiv.org/abs/2507.09288 (Local File: Paper_03_2507.09288_Hybrid_Quantum_Security_for_IPsec.pdf)

2. Abstract (Verbatim / Kept Same): Quantum Key Distribution (QKD) offers information-theoretic security against quantum computing threats, but integrating QKD into existing security protocols remains an unsolved challenge due to fundamental mismatches between pre-distributed quantum keys and computational key exchange paradigms. This paper presents the first systematic comparison of sequential versus parallel hybrid QKD-PQC key establishment strategies for IPsec, revealing fundamental protocol design principles that extend beyond specific implementations. We introduce two novel approaches for incorporating QKD into Internet Key Exchange version 2 (IKEv2) with support for both ETSI GS QKD 004 stateful and ETSI GS QKD 014 stateless API specifications: (1) a pure QKD approach that replaces computational key derivation with identifier-based quantum key coordination, and (2) a unified QKD-KEM abstraction that enables parallel composition of quantum and post-quantum cryptographic methods within existing protocol frameworks. Our key insight is that parallel hybrid approaches eliminate the multiplicative latency penalties inherent in sequential methods mandated by RFC 9370, achieving significant performance improvements under realistic network conditions. Performance evaluation using a Docker-based testing framework with IDQuantique QKD hardware demonstrates that the parallel hybrid approach significantly outperforms sequential methods under network latency conditions, while pure QKD achieves minimal bandwidth overhead through identifier-based key coordination. Our implementations provide practical quantum-enhanced IPsec solutions suitable for critical infrastructure deployments requiring defense-in-depth security.

3. Introduction: The increasing anticipation of quantum computing capabilities renders traditional cryptographic methods vulnerable, demanding the exploration of quantum-resistant alternatives for secure communication protocols. In response to this quantum threat, the cryptographic community has developed two primary defense strategies: Post-Quantum Cryptography (PQC) [1], which relies on mathematical problems believed to be hard even for quantum computers, and Quantum Key Distribution (QKD) [2], [3], which provides informationtheoretic security based on fundamental quantum mechanical principles. The recent standardization of ML-KEM (Module-Lattice-Based Key Encapsulation Mechanism) as FIPS 203 [4] by NIST in August2024establishestheprimarypost-quantumkey establishment standard that organizations will deploy for quantum-resistant communications. While ML-KEM provides computational security against quantum attacks, hybrid approaches that combine standardized post-quantum algorithms with QKD could offer defense-in-depth strategies particularly valuable for critical infrastructure deployments. IPsec (Internet Protocol Security) is a widely deployed secure communication protocol suite that provides security at the network layer, through two main protocols: Authentication Header (AH) and Encapsulating Security Payload (ESP), to handle data integrity, authentication, and encryption. These protocols require cryptographic keys to function, which are exchanged using the IKE (Internet Key Exchange) protocol. IPsec's architecture was first standardized in RFC 2401 [5], later updated in RFC 4301 [6], while key management evolved from IKEv1 [7] and ISAKMP (Internet Security Association and Key Management Protocol) [8] to the initialIKEv2specification[9], andfinallytoitscurrent form in RFC 7296 [10].

4. Literature Review & Findings: [Venue/Publisher: arXiv Computer Science - Cryptography and Security (cs.CR), 2025 | Methodology: Comparative protocol design and Docker-based strongSwan + IDQuantique hardware evaluation of sequential (RFC 9370) vs. parallel hybrid QKD-PQC in IKEv2. | Key Findings: Parallel hybrid QKD-KEM composition eliminates multiplicative RTT latency penalties of RFC 9370 sequential key exchanges under realistic network loss. | Research Gap: Requires specialized QKD-KEM plugins in strongSwan and lacks an automated analyzer to audit existing enterprise IPsec deployments for hybrid readiness.] 2 Related Work The integration of QKD with established security communication protocols has evolved significantly over the past two decades, with approaches targeting both IPsec and TLS frameworks. These developments have progressed from simple adaptations to sophisticated hybrid implementations. 2.1 QKD Integration with IPsec Early research by Sfaxi et al. [22] introduced SeQKEIP, which added a preliminary quantum key exchange phase to the ISAKMP framework, deliberatelybypassingIKEcompatibilityissues.

Berzanskis et al. [23] took a different approach with a hybrid model that combined classical IKE and QKDderived keys through XOR operations, requiring both systems to be compromised for a successful attack. The AQUA approach by Nagayama and Van Meter [19] proposed IKEv2 extensions for QKD integration, introducing QKD KeyID and Fallback payloads to exchange identifiers for pre-shared quantum keys rather than transmitting Diffie-Hellman values during key exchange. DARPA's quantum network implementation [24], analyzed by Dervisevic and Mehic [25], employed a 'rapid-reseeding' approach where QKD-derived bits supplemented the traditional IKEv1 key material in the KEYMAT hash function. This technique was further refined by MagiQ Technologies through direct XOR combination of QKD and IKE-generated keys. These early implementations highlighted key synchronization and rekeying rates as critical challenges for practical deployment. The European SECOQC project (2004-2008) demonstrated practical QKD-IPsec integration using VPN tunnels with AES encryption refreshed every 20 seconds via qu

Paper 04 [Sarvesh Bhamare]: Quantum-Ready Secure WAN: A Risk Assessment and Migration Framework

1. APA Reference: Alam, S. (2026). Quantum-Ready Secure WAN: A Risk Assessment and Migration Framework. arXiv preprint arXiv:2609.26225. https://arxiv.org/abs/2609.26225 (Local File: Paper_04_2609.26225_Quantum_Ready_Secure_WAN_A_Risk_Assessment_and_Mig.pdf)

2. Abstract (Verbatim / Kept Same): Enterprise wide-area networks (WANs) use quantum-vulnerable public-key cryptography to authenticate peers and establish keys for Internet Protocol Security (IPsec), Transport Layer Security (TLS), and software-defined WAN services. Harvest-now, decrypt-later collection already threatens data whose protection lifetime may exceed the time needed to migrate, while a future cryptographically relevant quantum computer would also endanger certificates and other authentication dependencies. Post-quantum standards and government directives now define algorithms and transition milestones, but they do not tell an enterprise which WAN services to migrate first or what evidence is sufficient for deployment. This paper proposes a vendor-neutral framework that separates exposure priority, migration readiness, and evidence confidence, and maps those outputs to staged security, operational, and governance gates. It also defines a complete evaluation design for classical and ML-KEM hybrid key establishment in TLS 1.3 and IKEv2/IPsec. The retained artifact is narrower: it contains author-reported aggregate TLS p50, p95, and p99 latencies at configured loss settings of 0%, 1%, 3%, and 5%, an arithmetic checker, and limited environment and MTU/MSS notes. At those settings, the TLS-H minus TLS-C p95 differences are 0.00, 0.16, 0.38, and 0.63 ms, corresponding to 0.00%, 1.32%, 3.07%, and 5.00% relative to TLS-C. These values are descriptive aggregate observations, not inferential results, because per-trial data, sample counts, original captures, and timestamped logs were not retained. Protocol-size and MTU/MSS calculations are reported separately from measured behavior, and unsupported IKE, resource, throughput, retransmission, and downgrade outcomes are excluded.

3. Introduction: Enterprise wide-area networks (WANs) carry business traffic between headquarters, branch offices, data centers, cloud environments, remote users, and external partners. These connections commonly use IPsec/IKEv2, TLS, and related public-key mechanisms to authenticate endpoints and establish symmetric session keys. Their security depends heavily on RSA, finite-field Diffie-Hellman, and elliptic-curve cryptography. These algorithms remain suitable against known classical attacks when correctly configured. However, a sufficiently capable quantum computer could solve the integer-factorization and discrete-logarithm problems on which their security depends [1]. Such a system - generally described as a cryptographically relevant quantum computer (CRQC) - is not known to exist, and its arrival date remains 2 uncertain. The uncertainty does not remove the risk because cryptographic migration and data exposure operate on longer timelines. An adversary does not need a CRQC today to create future harm. In a harvest-now, decrypt-later attack, encrypted traffic is collected and retained until a quantum computer or another cryptanalytic advance makes decryption practical. The immediate risk therefore depends not only on when a CRQC becomes available, but also on how long the intercepted information must remain confidential and how long the organization will need to replace vulnerable cryptography. If the required confidentiality period and migration time together extend beyond the estimated time to a CRQC, delayed migration can expose data before replacement is completed [2]. This condition is especially relevant to WAN traffic containing intellectual property, regulated personal information, authentication material, financial records, strategic plans, or government-controlled data.

4. Literature Review & Findings: [Venue/Publisher: arXiv Computer Science - Cryptography and Security (cs.CR), 2026 | Methodology: Vendor-neutral risk assessment and staged governance framework evaluating exposure priority, migration readiness, and ML-KEM hybrid IKEv2/TLS latency. | Key Findings: Formalizes the harvest-now-decrypt-later risk inequality (L + M > Q) and quantifies MTU/MSS fragmentation and p95 latency overheads across 0-5% packet loss. | Research Gap: Provides governance gates and TLS/IKE analytical models but lacks automated packet-capture-based cryptographic inventory and AI risk scoring.] 3 Related Work and Research Gap Research relevant to quantum-ready WAN migration spans organizational transition, cryptographic discovery, cryptographic agility, protocol design,

measurement, and risk communication. The framework adopts established work in each stream and focuses on the unresolved step of joining those inputs at the WAN-service level. 3.1 Organizational Migration and Cryptographic Agility Joseph et al. frame post-quantum migration as a multi-year organizational problem involving inventories, standards, suppliers, applications, and governance [2]. Ott et al. identify related crypto-agility research challenges [10], while Hasan et al. use dependency analysis to expose relationships that a flat inventory misses [17]. NIST's crypto-agility guidance extends the scope 9 across protocols, software, hardware, firmware, and infrastructure [19]. This paper adopts that dependency-oriented view and narrows the assessment unit to an operational WAN service. Their main limitation for the present problem is granularity: none defines how a WAN operator should keep exposure priority, deployment preparation, and evidence quality separate when comparing individual tunnels or application relationships. 3.2 Cryptographic Discovery and Dependency Mapping The NIST NCCoE migration project examines cryptographic discovery and inventory construction [18]. Configuration analysis, packet inspection, software inventories, and certificate records can show where quantum-vulnerable cryptography is used. The framework uses these outputs as evidence, then adds service ownership, timing, imp

Paper 05 [Sarvesh Bhamare]: AWA: Adversarial Website Adaptation

1. APA Reference: Sadeghzadeh, A. M., Tajali, B., & Jalili, R. (2020). AWA: Adversarial Website Adaptation. arXiv preprint arXiv:2012.10832. https://arxiv.org/abs/2012.10832 (Local File: Paper_05_2012.10832_AWA_Adversarial_Website_Adaptation.pdf)

2. Abstract (Verbatim / Kept Same): One of the most important obligations of privacy-enhancing technologies is to bring confidentiality and privacy to users' browsing activities on the Internet. The website fingerprinting attack enables a local passive eavesdropper to predict the target user's browsing activities even she uses anonymous technologies, such as VPNs, IPsec, and Tor. Recently, the growth of deep learning empowers adversaries to conduct the website fingerprinting attack with higher accuracy. In this paper, we propose a new defense against website fingerprinting attack using adversarial deep learning approaches called Adversarial Website Adaptation (AWA). AWA creates a transformer set in each run so that each website has a unique transformer. Each transformer generates adversarial traces to evade the adversary's classifier. AWA has two versions, including Universal AWA (UAWA) and Non-Universal AWA (NUAWA). Unlike NUAWA, there is no need to access the entire trace of a website in order to generate an adversarial trace in UAWA. We accommodate secret random elements in the training phase of transformers in order for AWA to generate various sets of transformers in each run. We run AWA several times and create multiple sets of transformers. If an adversary and a target user select different sets of transformers, the accuracy of adversary's classifier is almost 19.52% and 31.94% with almost 22.28% and 26.28% bandwidth overhead in UAWA and NUAWA, respectively. If a more powerful adversary generates adversarial traces through multiple sets of transformers and trains a classifier on them, the accuracy of adversary's classifier is almost 49.10% and 25.93% with almost 62.52% and 64.33% bandwidth overhead in UAWA and NUAW, respectively.

3. Introduction: W ebsite fingerprinting (WF) attack is one of the most serious threats against the anonymity of the users’ browsing activities. It enables an adversary to determine which website is visited by a target user even she uses Privacy Enhancing Technologies (PETs), such as VPNs, IPsec, and Tor [1], [2], [3], [4], [5], [6], [7]. The target user is a general term referring to a client-side victim whom the adversary intends to monitor and figure out which specific websites she has been visiting through her browser application. When an adversary visits a website through PETs, she can extract a unique trace associated with the statistical features of that website’s network flow, such as the sequence of packets’ direction. The adversary can visit various websites several times and collect a trace set from those. To conduct the website fingerprinting attack, the adversary can train a classifier on the collected trace set and uses this classifier to predict the website that has been visited by the target user. The fundamental part of website fingerprinting attack is the classifier being used by the adversary to classify the target user’s browsing activities. Recently, Deep The authors are with the Department of Computer Enginnering, Sharif University of Technology, Iran, Tehran, 11365-11155 (e-mail: amsadeghzadeh@ce.sharif.edu; behradtajali@ce.sharif.edu; jalili@sharif.edu). Neural Networks (DNNs) have shown great performance in classifying network traces of various websites[5], [6]. Besides, DNNs-based classifiers have a highly better performance on defenses that has been proposed in the previous studies against website fingerprinting attack, such as WTF-PAD[5]. However, researchers have recently shown that DNNs have a serious vulnerability, which is called adversarial example [8].

4. Literature Review & Findings: [Venue/Publisher: IEEE Transactions on Information Forensics and Security (TIFS), IEEE | Methodology: Adversarial Website Adaptation (AWA) using deep learning transformer sets (UAWA and NUAWA) with secret random elements to perturb VPN/IPsec traffic traces. | Key Findings: Reduces deep learning website fingerprinting attacker accuracy to 19.52% (UAWA) and 25.93% (NUAWA) even against adversarially trained classifiers. | Research Gap: Focuses on defending user web browsing privacy against traffic classifiers rather than auditing VPN

protocol security configurations.] II. BACKGROUND The Onion Router (Tor) is one of the most popular PETs, which provides anonymity and protection from eavesdropping. Different kinds of research have been conducted previously to breach the anonymity feature of Tor [1], [2], [3], [4], [5], [6], [7]. Website fingerprinting is one such type of traffic analysis attack through extracting traffic patterns from the visited websites can form a unique fingerprint of acquired traffic flow and undermine Tor protection by detecting the website that the user has been just visiting. Tor Client Adversary Tor Network Bridge Node Middle Nodes Exit Node Internet Tor Encrypted Traffic Figure 1: A depiction for a typical WF attack scenario. A. Website Fingerprinting (WF) Three entities play vital roles in the website fingerprinting attack, an adversary, a target user, and Privacy-Enhancing Technologies (PETs). A target user utilizes PETs to keep her communications confidential and anonymous. Since most PETs are publicly available, an adversary can use them to generate a set of network flows from her favorite websites and train a classifier on them to predict the target user's browsing activities. AS adversary has no access to the label of network flows being generated by the target user, she can not evaluate the performance of her classifier. A website fingerprinting defense mechanism aims to obfuscate the features from which the adversary's classifier learns to recognize and differentiate network flows of a website from those of others. Likewise the previous studies, in this paper, we focus on Tor as the most popular PETs. Ho

Paper 06 [Sarvesh Bhamare]: Q-ESP: a QoS-compliant Security Protocol to enrich IPSec Framework

1. APA Reference: Mostafa, M., Kalam, A. A. E., & Fraboul, C. (2009). Q-ESP: a QoS-compliant Security Protocol to enrich IPSec Framework. Third IEEE / IFIP International Conference on New Technologies, Mobility and Security, Cairo : Egypt (2009). https://arxiv.org/abs/0911.4034 (Local File: Paper_06_0911.4034_Q_ESP_a_QoS_compliant_Security_Protocol_to_enrich.pdf)

2. Abstract (Verbatim / Kept Same): IPSec is a protocol that allows to make secure connections between branch offices and allows secure VPN accesses. However, the efforts to improve IPSec are still under way; one aspect of this improvement is to take Quality of Service (QoS) requirements into account. QoS is the ability of the network to provide a service at an assured service level while optimizing the global usage of network resources. The QoS level that a flow receives depends on a six-bit identifier in the IP header; the so-called Differentiated Services code point (DSCP). Basically, Multi-Field classifiers classify a packet by inspecting IP/TCP headers, to decide how the packet should be processed. The current IPSec standard does hardly offer any guidance to do this, because the existing IPSec ESP security protocol hides much of this information in its encrypted payloads, preventing network control devices such as routers and switches from utilizing this information in performing classification appropriately. To solve this problem, we propose a QoS-friendly Encapsulated Security Payload (Q-ESP) as a new IPSec security protocol that provides both security and QoS supports. We also present our NetBSD kernel-based implementation as well as our evaluation results of Q-ESP.

3. Introduction: In today's business environments, users demand seam less connectivity and stable access to servers and netwo rks wherever they are: hotels, airports, remote offices , etc. While these functionalities are useful for business, they work only if we can minimize the security risks of transferring sensitive data across the Internet. In order to achieve this goal, there are various security mechanisms for network environment ; the most popular is Security Architecture for IP (IPSec) [1]. IPSec is designed to provide inter-operable cryptographic ally-based security for IPv4 and lPv6. IPSec operates at the I P layer, making it transparent to applications and users. Unfortunately, security does not come for free and, in general, protection mechanisms require more process ing time and causes traffic delay. Real-time applications su ch as video conferencing, VoIP, and real-time video need specia l processing to achieve their goals and to overcome t he delay introduced by adding security mechanisms. Quality o f service (QoS) has been emerged to solve a part of this prob lem by providing priority treatment to real time traffic. In the QoS domain, the Class of Service concept divides the ne twork traffic into different classes and provides a class -dependent service to each packet. To classify packets, each p acket is assigned a priority value. The latter is stored in the "Type of Service" (ToS) field in the IP v4 header (also call ed Traffic Class in IP v6). However, allowing the sending device to classify tr affic or to set traffic priorities may be subject to threats , as the sender may classify his traffic in a way that gives him upper priorities. This is clearly the disadvantage of what is called passive admission control . Conversely, service providers perform active admission

4. Literature Review & Findings: [Venue/Publisher: 3rd IEEE/IFIP International Conference on New Technologies, Mobility and Security (NTMS), IEEE | Methodology: Design and NetBSD kernel implementation of QoS-friendly Encapsulated Security Payload (Q-ESP) exposing controlled flow classification fields. | Key Findings: Enables edge routers to perform Multi-Field Classification (MFC) and enforce QoS priority for real-time VoIP/video over IPsec tunnels with <0.1% throughput overhead. | Research Gap: Exposing additional header metadata for QoS classification increases side-channel traffic fingerprinting exposure if not monitored by an intelligent security analyzer.] curity protocol. IPSec ESP protocol encrypts the transport layer header, and thus hides the source and destina tion port numbers as

well as the protocol identifier. To solve this problem, we introduce the Q-ESP " QoS friendly Encapsulated Security Payload " protocol that enforces both QoS and security requirements. The ma jor aim of our Q-ESP is to construct packets that are QoS c ontrollable according to active admission control while providi ng the same security services ensured by IPSec ESP and AH [6]. The remainder of this paper is organized as follows : Section II presents our Q-ESP protocol packet structure. Th en, Section III gives the details of Q-ESP processing. Section IV focuses on Q-ESP implementation. Afterward, Section V, prov ides our Q-ESP analytical evaluation. In Section VI, we pres ent the performances evaluation experiments and results. Fi nally, Section VII draws up conclusions and future works. Fig. 1. The AH format. Fig. 1. The ESP format. Fig. 3. ESP in transport mode and tunnel mode. Fig. 4. Q-ESP packet format in Ipv4. Fig. 5. Q-ESP packet format in Ipv6. Figure 6. Q-ESP implementation. Fig. 7. Testbed for throughput experiments. . Figure 8. Throughput using different (encryption/authentication) algorithms for ESP and Q-ESP TABLE I: Q-ESP AND ESP THROUGHPUT (K BPS). Packet size ESP Q-ESP 64 51.243 51.191 128 102.366 102.366 256 204.715 204.834 512 409.600 409.463 1024 819.268 818.654 2048 1638.127 1637.444 4096 3275.435 3275.162 Fig. 8. Testbed for priority control experiments. 0 50 100 150 200 250 300 350 400 450 100 200 300 400 500 600 700 800 9

Paper 07 [Sarvesh Bhamare]: P4-IPsec: Site-to-Site and Host-to-Site VPN with IPsec in P4-Based SDN

1. APA Reference: Hauser, F., Häberle, M., Schmidt, M., & Menth, M. (2019). P4-IPsec: Site-to-Site and Host-to-Site VPN with IPsec in P4-Based SDN. arXiv preprint arXiv:1907.03593. https://doi.org/10.1109/ACCESS.2020.3012738 (Local File: Paper_07_1907.03593_P4_IPsec_Site_to_Site_and_Host_to_Site_VPN_with_IP.pdf)

2. Abstract (Verbatim / Kept Same): In this work, we present P4-IPsec, a concept for IPsec in software-defined networks (SDN) using P4 programmable data planes. The prototype implementation features ESP in tunnel mode and supports different cipher suites. P4-capable switches are programmed to serve as IPsec tunnel endpoints. We also provide a client agent to configure tunnel endpoints on Linux hosts so that site-to-site and host-to-site application scenarios can be supported which are the base for virtual private networks (VPNs). While traditional VPNs require complex key exchange protocols like IKE to set up and renew tunnel endpoints, P4-IPsec benefits from an SDN controller to accomplish these tasks. One goal of this experimental work is to investigate how well P4-IPsec can be implemented on existing P4 switches. We present a prototype for the BMv2 P4 software switch, evaluate its performance, and publish its source code on GitHub. We explain why we could not provide a useful implementation with the NetFPGA SUME board. For the Edgecore Wedge 100BF-32X Tofino-based switch, we presented two prototype implementations to cope with a missing crypto unit. As another contribution of this paper, we provide technological background of P4 and IPsec and give a comprehensive review of security applications in P4, IPsec in SDN, and IPsec data plane implementations. According to our knowledge, P4-IPsec is the first implementation of IPsec for P4-based SDN.

3. Introduction: Virtual Private Networks (VPNs) extend private networks across public networks by adding authentication and encryption to network traffic. Internet Protocol Security (IPsec) is one of the oldest, but still most-widespread VPN protocols. Standardized by the IETF, it introduces protection on the Internet Protocol (IP) layer. Due to its large distribution, many implementations for network appliances and operating systems are available. Although it is criticized for its complexity, proven deployment patterns allow efficient and reliable operation. IPsec tunnel setup requires user configuration plus keying material that is exchanged by IPsec peers via the Internet Key Exchange (IKE) protocol. Complexity grows with the number of IPsec peers, especially in highly dynamic environments such as campus or enterprise networks with many users and sites. Several works investigate on how to leverage the centralized control plane of software-defined networking (SDN) Frederik Hauser, Marco Häberle, Mark Schmidt, and Michael Menth are with Chair of Communication Networks, University of Tuebingen, Tuebingen, Germany. E-mail: {frederik.hauser,marco.haeberle,markthomas.schmidt,menth}@uni-tuebingen.de This work was supported by the Deutsche Forschungsgemeinschaft (DFG) under grant ME2727/1-2 and the bwNET100G+ project which is funded by the Ministry of Science, Research and the Arts Baden-Württemberg (MWK). The authors alone are responsible for the content of this paper. to simplify IPsec operation. However, the possibilities for IPsec deployment in SDN were limited. Typical SDN switches have a fixed-function data plane that does not provide support for IPsec. As a result, IPsec data plane processing needs to be moved to an additional software-based packet processing function (PPF). Besides

4. Literature Review & Findings: [Venue/Publisher: IEEE Access (Vol. 8, 2020), IEEE | Methodology: Programmable data-plane implementation of IPsec ESP tunnel/transport modes in P4 switches (BMv2, NetFPGA, Edgecore Tofino) managed by an SDN controller. | Key Findings: Replaces complex distributed IKE key exchange with centralized gRPC SDN control, achieving scalable site-to-site and host-to-site IPsec VPN management. | Research Gap: Centralized SDN key distribution creates a single point of compromise if controller-to-switch channels or SPD/SAD policies are misconfigured.] related work in that field. We present an IPsec data plane implementation that integrates IPsec

components and processes with constructs and components under the given constraints of the P4 data plane programming language. Cryptographic operations for authentication, encryption, and decryption are implemented in P4 externs where IPsec components such as the Security Policy Database (SPD) and Security Association Database (SAD) are part of the P4 processing pipeline. P4 switches that implement the functionality of P4-IPsec can be deployed in host-to-site and site-to-site VPN scenarios. Control plane functions for IPsec operation are part of a central SDN controller that maintains IPsec tunnels without the help of distributed key exchange protocols such as IKE. As these components are steered by a centralized control plane through an authenticated and encrypted control connection, complex IKE-based key exchange protocols are substituted by controller-based tunnel setup and renewal procedures. For host-to-site operation, we introduce a client agent for Linux operating systems that runs on the roadwarrior hosts. It establishes an interface to the central SDN controller via a gRPC connection. To investigate how well P4-IPsec can be implemented on existing P4 targets, we work on three prototypes. We successfuly implement a prototype for the Behavioral Model version 2 (BMv2) P4 software target and conduct a performance evaluation. We release the source code of our prototype along its testbed environment under the Apache v2 license on GitHub [1]. In addition, we report on implement

Paper 08 [Sarvesh Bhamare]: Wireguard: An Efficient Solution for Securing IoT Device Connectivity

1. APA Reference: Jumakhan, H., & Mirzaeinia, A. (2024). Wireguard: An Efficient Solution for Securing IoT Device Connectivity. arXiv preprint arXiv:2402.02093. https://arxiv.org/abs/2402.02093 (Local File: Paper_08_2402.02093_Wireguard_An_Efficient_Solution_for_Securing_IoT_D.pdf)

2. Abstract (Verbatim / Kept Same): The proliferation of vulnerable Internet-of-Things (IoT) devices has enabled large-scale cyberattacks. Solutions like Hestia and HomeSnitch have failed to comprehensively address IoT security needs. This research evaluates if Wireguard, an emerging VPN protocol, can provide efficient security tailored for resource-constrained IoT systems. We compared Wireguards performance against standard protocols OpenVPN and IPsec in a simulated IoT environment. Metrics measured included throughput, latency, and jitter during file transfers. Initial results reveal Wireguard's potential as a lightweight yet robust IoT security solution despite disadvantages for Wireguard in our experimental environment. With further testing, Wireguards simplicity and low overhead could enable widespread VPN adoption to harden IoT devices against attacks. The protocols advantages in setup time, performance, and compatibility make it promising for integration especially on weak IoT processors and networks.

3. Introduction: With the general increase in smaller smart devices connected to the Internet, the IoT devices, there has also been a large increase in the risks inherent to their portability and connectivity. For example, in 2016, thousands of IoT devices were infected and then used as part of a botnet for a DDoS attack that targeted Twitter, Spotify, Paypal, and made unavailable other popular websites like CNN, New York Times, etc. [1]. The Mirai botnet, which used default password lists to break into vulnerable IoT devices, is a great example of a large-scale attack that was discovered in 2016 [2]. More recently, the Verkada hack in March 2021 was another major event with serious complications for user privacy as a result of IoT security failure [1]. From the above real-world examples, the possibility of using millions of unprotected IoT devices for such malicious attacks, as mentioned above, is quite visible. There are also other types of attacks that are possible with such devices. A highly sensitive medical or military facility’s connected devices or smart thermostats could be hijacked to coerce people into what the hackers demand [3]. We will discuss and then test the most popular secure tunneling solutions available that can be potentially used for securing IoT devices on the Internet. These include: 1) Widely used general-purpose security solutions such as Secure Socket Layer/Transport Layer Security (SSL/TLS) protocol that works at the application layer of the OSI Model to provide secure browsing on almost all web pages. 2) Virtual Private Networks that work at the network or transport layer and also provide an encrypted tunnel, albeit with properties different from those of the SSL/TLS protocol. A detailed comparison of VPN and SSL/TLS is to follow after discussing two other

4. Literature Review & Findings: [Venue/Publisher: International Conference on Computational Science and Computational Intelligence (CSCI), IEEE | Methodology: Experimental benchmarking of WireGuard, IPsec, and OpenVPN across throughput, latency, and jitter metrics in constrained IoT network environments. | Key Findings: Demonstrates tradeoffs between IPsec's extensive cipher suite flexibility and WireGuard's fixed modern primitives (ChaCha20-Poly1305) on weak processors. | Research Gap: IPsec's extensive configurability (legacy ciphers, multiple modes) creates severe misconfiguration risks compared to fixed-cipher protocols, motivating automated IPsec auditing.] II. RELATED WORK AND BACKGROUND A. Hestia Hestia tries to prevent the hijacking of devices on a local network by only allowing controller devices like a Google Home module, an Amazon Alexa, etc. to connect to controlled devices like cameras, door locks, smart lights, and such. Below are some issues with Hestia that render it unreliable from a security and ease of usability perspective: 1) The controller devices can themselves have unreliable protection and can possibly be hijacked by an attacker. This happens when controllers are not updated and/or they have a vulnerability,

which can be extremely difficult to cover since there are numerous manufacturers and they each have many more products. Since the controllers have access to all devices on the network, it is still risky to rely on such a setting. 2) With blocking the controlled devices from the Internet completely, the administrator/user creates a dependency on the controller devices. That is, the user has to be able to control the controller devices to control the controlled devices at the premise. This could increase the risk of damage to property if and when the controllers are hijacked and the owner is at a remote location since he/she is shut off from all the controlled devices by design. B. HomeSnitch HomeSnitch is another proposed solution, and it tries to learn normal device behavior for every IoT at a premise over a certain period of normal use and then effectively firewalls packets that have not routinely been sent or received in hopes of preventing and busting the hacking of these devices. Although it is an e

Paper 09 [Sarvesh Bhamare]: Study of Post Quantum status of Widely Used Protocols

1. APA Reference: Mallick, T., Kundu, A., & Kompella, R. (2026). Study of Post Quantum status of Widely Used Protocols. arXiv preprint arXiv:2603.28728. https://arxiv.org/abs/2603.28728 (Local File: Paper_09_2603.28728_Study_of_Post_Quantum_status_of_Widely_Used_Protoc.pdf)

2. Abstract (Verbatim / Kept Same): The advent of quantum computing poses significant threats to classical public-key cryptographic primitives such as RSA and elliptic-curve cryptography. As many critical network and security protocols depend on these primitives for key exchange and authentication, there is an urgent need to understand their quantum vulnerability and assess the progress made towards integrating post-quantum cryptography (PQC). This survey provides a detailed examination of nine widely deployed protocols - TLS, IPsec, BGP, DNSSEC, SSH, QUIC, OpenID Connect, OpenVPN, and Signal Protocol - analysing their cryptographic foundations, quantum risks, and the current state of PQC migration. We find that TLS and Signal lead the transition with hybrid post-quantum key exchange already deployed at scale, while IPsec and SSH have standardised mechanisms but lack widespread production adoption. DNSSEC and BGP face the most significant structural barriers, as post-quantum signature sizes conflict with fundamental protocol constraints. Across all protocols, key exchange proves consistently easier to migrate than authentication, and protocol-level limitations such as message size and fragmentation often dominate over raw algorithm performance. We also discuss experimental deployments and emerging standards that are shaping the path towards a quantum-resistant communication infrastructure.

3. Introduction: Quantum computing uses quantum mechanics to solve certain problems exponentially faster than classical computers. In particular, Shor’s algorithm breaks the integer factorisation and discrete logarithm problems that underpin RSA, Diffie–Hellman, and ellipticcurve cryptography. Although large-scale quantum computers do not yet exist, adversaries can already record encrypted traffic today and decrypt it once quantum capabilities mature —the so-called “harvest-now, decrypt-later” threat. For data requiring long-term confidentiality, this renders current protections inadequate well before a quantum computer is physically built. NIST conducted a multi-year effort to standardise quantumresistant algorithms, releasing public drafts in 2023 [53, 58–60] and publishing the first PQC standards —FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) —in August 2024 [54–57]. In ∗This work was carried out when the author was at Cisco Research, San Jose, CA, during Summer Internship 2025. He is currently at Northeastern University, Boston, MA. Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org. Conference acronym ’XX, Woodstock, NY ©2018 Copyright held by the owner/author(s). Publication rights licensed to ACM. ACM ISBN 978-1-4503-XXXX-X/18/06 https://doi

4. Literature Review & Findings: [Venue/Publisher: ACM / arXiv Cryptography and Network Security Survey, 2026 | Methodology: Systematic cross-protocol security audit of 9 core protocols (TLS, IPsec, SSH, BGP, DNSSEC, QUIC, OpenVPN, OIDC, Signal) for Post-Quantum Cryptography (PQC). | Key Findings: Reveals that while IKEv2 has RFC 9242 (IKE_INTERMEDIATE) and RFC 9370 standards, production IPsec VPNs lag behind TLS in hybrid PQC deployment due to UDP fragmentation. | Research Gap: Highlights urgent need for automated protocol analyzers that detect quantum-vulnerable DH groups (Groups 1, 2, 5, 14) and guide RFC 9370 migration.] 2 Background PQ cryptography is crucial to ensuring security in the face of emerging quantum computing threats. As quantum computers advance, they pose a risk to classical cryptographic algorithms, making the adoption of PQ key exchange mechanisms and digital signature schemes essential. Gradually, these PQ algorithms are being integrated into existing protocols to enhance security and future-proof systems against quantum attacks. Below we overview current PQ algorithms and some protocols that implement them. 2.1 KEM and Digital Signature Algorithms The algorithms are categorized based on the mathematical problems they rely on, each offering different strengths, weaknesses, and Conference acronym

’XX, June 03–05, 2018, Woodstock, NY Tushin Mallick, Ashish Kundu, and Ramana Kompella applications. A summary of the algorithms is provided in Table 1 and we expound on the categories below. Lattice-basedcryptography is a leading class of PQC, grounded in hard lattice problems such as the Shortest Vector Problem and Learning With Errors, which are believed to resist both classical and quantum attacks. These schemes offer favorable efficiency and compactness, making them well suited for key exchange, encryption, and digital signatures. Notable examples include the key-exchange mechanisms CRYSTALS-Kyber [14], NTRU [39], and SABER [22], as well as the digital signature scheme CRYSTALS-Dilithium, all of which are finalists in the NIST PQC standardization process. Code-basedcryptography is rooted in the hardness of decoding random linear codes—a problem considered difficult even for qu

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